

Today

Runge-Kutta

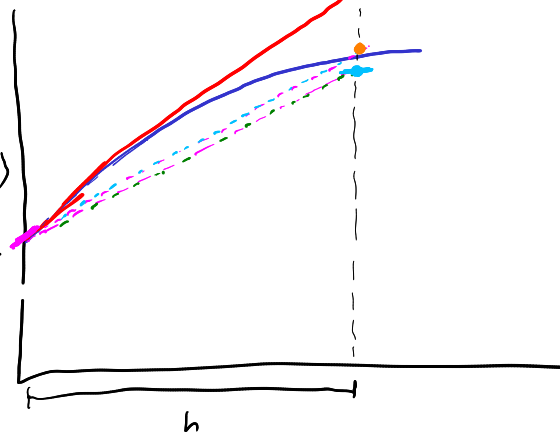
Heun's algorithm

Step i:

predict: $x_{i+1}^0 \leftarrow x_i^m + f(t_i, x_i)h$

Repeat until $\left| \frac{x_{i+1}^{j+1} - x_{i+1}^j}{x_{i+1}^j} \right| < \epsilon_s$: (n iterations over j)

correct: $x_{i+1}^j \leftarrow x_i^m + h \frac{f(t_i, x_i^m) + f(t_{i+1}, x_{i+1}^{j-1})}{2}$
 $j \leftarrow j+1$



Local: $O(h^3)$ Global: $O(h^2)$

Runge Kutta

Classical RK4 "4th order Runge Kutta"

update

$$x_{i+1} \leftarrow x_i + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)h$$

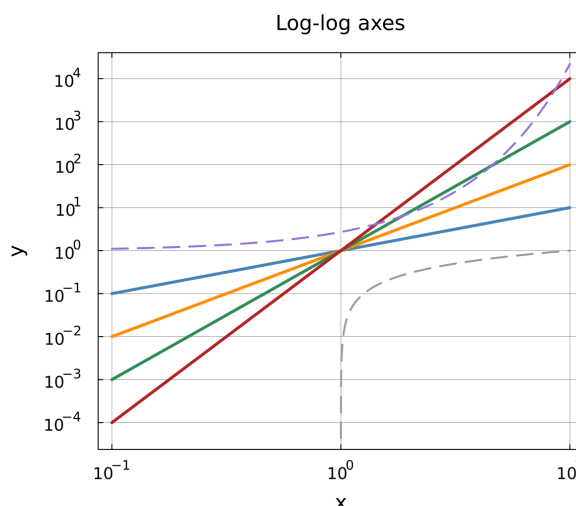
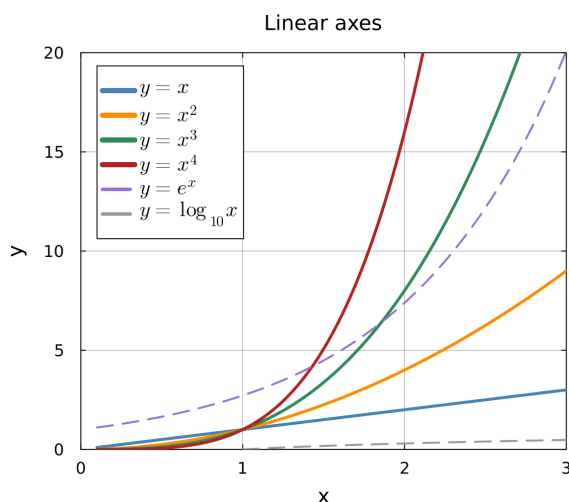
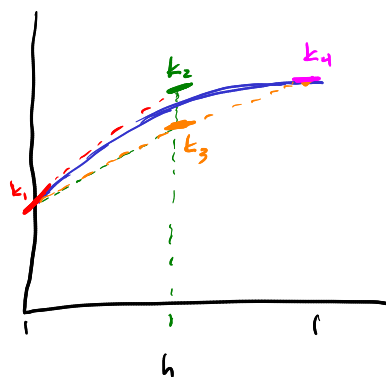
$$k_1 = f(t_i, x_i)$$

$$k_2 = f(t_i + \frac{1}{2}h, x_i + \frac{1}{2}k_1h)$$

$$k_3 = f(t_i + \frac{1}{2}h, x_i + \frac{1}{2}k_2h)$$

$$k_4 = f(t_i + h, x_i + k_3h)$$

Local: $O(h^5)$ Global: $O(h^4)$



coordinates on log-log

$$x' = \log(x) \quad y' = \log(y)$$

$$y' = \log(x^k) = k \log(x) = kx'$$

$$y' = kx'$$

$y = x^k$

Announcements

Lab 2, HW4 due Thurs

Exam Grades today hopefully

General Runge-Kutta

$$x_{i+1} \leftarrow x_i + \phi h$$

$$\phi = a_1 k_1 + a_2 k_2 + \dots + a_n k_n$$

$$k_1 = f(t_i, x_i)$$

$$k_2 = f(t_i + p_1 h, x_i + q_{11} k_1 h)$$

$$k_3 = f(t_i + p_2 h, x_i + q_{21} k_1 h + q_{22} k_2 h)$$

$\vdots$

$$k_n = f(t_i + p_{n-1} h, x_i + q_{n-1,1} k_1 h + \dots + q_{n-1,n-1} k_{n-1} h)$$

How to find a 's, p 's, q 's?

Set equal to order- n Taylor Series

Example: Second Order

$$a_1 + a_2 = 1$$

$$a_2 p = \frac{1}{2}$$

$$a_2 q = \frac{1}{2}$$

3 equations
4 unknowns

under-determined

① $a_2 = \frac{1}{2} \Rightarrow a_1 = \frac{1}{2}, p = q = 1$

Heun's with no iter

② $a_2 = 1 \Rightarrow a_1 = 0, p = q = \frac{1}{2}$

Midpoint

③ $a_2 = \frac{3}{4} \Rightarrow a_1 = \frac{1}{4}, p = q = \frac{2}{3}$

Ralston's